

 सत्यमेव जयते	भारत सरकार दूरसंचार विभाग दूरसंचार अभियांत्रिकी केंद्र Government of India Department of Telecommunications Telecommunication Engineering Centre K.L. Bhawan, Janpath, New Delhi- 110001	 ISO 9001:2015
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No. 26-1/2022-FA/TEC

Date: 05.06.2023

Subject: Minutes of sixth meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group (SG)-15 held on 25.05.2023.

The sixth meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group 15 regarding “*Networks, Technologies and Infrastructures for Transport, access & home*”, reconstituted for the Study Period 2022-24, was held on 25.05.2023 under the chairpersonship of Sh. Rakesh Desai, DDG(FA), TEC through video-conferencing. List of participants is placed at Annexure-A.

2. At the outset, Sh. Deepak Pathak, Director (FA) TEC & Convener of NWG-15, welcomed all the participants and briefly explained about the discussions held during the previous meetings. He briefly informed achievements of NWG15 in submitting contributions at ITU-T SG15 leading to be part of the final recommendations or the final recommendation of ITU-T SG15. He congratulated Shri Sudipta Bhaumik, STL & Vice-Chair of SG15 on his selection as Vice Chair of Working Party 2 of SG15 during last meeting of SG15 held at Geneva from 17-28 April 2023.

3. Sh. Rakesh Desai, Chairman NWG-15 & DDG (FA) TEC, greeted all the members and specially welcomed the new members. He informed that new members from academia, industry/association have joined NWG-15. He highlighted that three contributions have been submitted by India in last meeting of ITU-T SG15 and further informed that the next meeting of ITU-T SG15 is scheduled from 20th Nov to 1st Dec 2023 at Geneva.

4. After the introduction, Sh. Rakesh Desai explained the outcome of the recently concluded SG15 meeting as under: -

- (i) The SG15 appreciated the contribution (Q6/15) regarding “System, Functional and Engineering requirements for establishing Free Space Optical (FSO) transmission systems” and requested further contributions listing the different use cases of FSO systems discussing system requirements, technology maturity and applicability in multi-vendor interoperable scenarios for each case. The contributors of this work-item have been requested to prepare afresh contribution and other members of the NWG-15 are requested to give their inputs in this subject. (Action By the contributors: Abdul Kayum/ Avadhesh Singh/ Rakesh Goyal / Rakesh Desai)
- (ii) Contribution (Q3/15) proposing to include use cases of Li-fi in healthcare and education areas in the ‘Technical Paper on the use of ITU-T G.9991 technology for Light Communication (LC)’ has been accepted for including as working text in the technical paper on G.9991 technology. Indian Contribution may be further updated at the time of finalization of the said technical paper if required. (Action By the contributors: Rakesh Desai/ Mudita Chandra/ Hardik Soni, M/s NavTech / Deepak Solanki, Velminni)
- (iii) Contribution (Q7/15) on “Topologies for optical access network” was also accepted by

the SG15 wherein the Indian experience in the areas of optical access network topologies has been explained. The SG15 meeting agreed to start a new L-series Supplement, L.Sup.fttx, "National Experiences for FTTx Network Architectures" to cover all national experience as information for Recommendation L.250. It was further agreed to include Indian experience into this new supplement. India has been listed as one of the supporting members that are committing to contributing actively to this work item. Indian Contribution may be further updated by the contributors at the time of finalization of the said new supplement L.Sup.fttx if required and inputs are invited from other members of the NWG-15 (Action By the contributors: Rakesh Desai/ Mudita Chandra/ BSNL).

- (iv) SG15 decided, in the framework of the studies conducted under Q7/15, to produce a Questionnaire to collect detailed background information on the issues related to outdoor optical infrastructures sharing. This Questionnaire is regarding practical and technical aspects of the outdoor optical infrastructures sharing between multiple operators. Ms. Mudita, ADET, TEC under the guidance of Sh. Rakesh Desai is working on this work-item. Members of the NWG-15 are requested to give their inputs by **15th July 2023** in respect of this work-item. (Action By: Rakesh Desai / Mudita)
- (v) SG15 (Q5 to 8) is working on proposed draft revised ITU-T G.Sup.40 "Optical fibre and cable Recommendations and standards guideline" and for this, correspondence activity to update of ITU-T Technical Paper LSTP-GLSR "Guide on the use of ITU-T L-series Recommendations related to optical technologies for outside plant" has been already initiated by the Rapporteur. The NWG-15 members from Sterlite, HFCL, BSNL, DOT LSAs are requested to give their inputs by **13th June 2023**.

5. Sh. Rakesh Desai also apprised the participants about the members who actively contributed during the SG15 study period 2022-24. He mentioned that India's contribution in ITU-T is minimal and need to work on more areas. He emphasized on possible areas for contributions like use of AI/ML in networking planning, Network automation, Asymmetric splitting in optical access network, Subnetwork management through OLT, Support of interface such as eCPRI, Improvement in ONU activation method, Implementation of PQC/QKD, Amendment to ITU recommendation L.100, Revision of G.Sup.40, Revision of G.Sup.650.1, Revision of G.Sup.654, Revision of L.109 etc.

6. Sh. Rakesh Desai appreciated the response of the NWG-15 members in registering on 'Standards Coordination Portal' of TEC (<https://tec.gov.in/scp/>) and requested the remaining members to register and start familiarizing with the features of the portal. He further mentioned that important circulars, meeting notices and minutes of meeting will be now onwards posted on this portal.

7. Sh. Abdul Kayum, DDG, while referring to the Indian contribution and MIIT, China work-item on FSOC (Q6/15), suggested that efforts may be initiated to merge our contribution to the agreed work-item of MIIT, China. He further suggested that the Rapporteur of Q6/15 may be interacted in this regard.

8. Sh. Abdul Kayum, DDG, suggested that 6G Division of TEC may take a lead in preparing the contribution (Q3/15) regarding PQC/QKD implementation to make the system secure in the era of quantum computing. (Action By: Sh. Raju / Rakesh Goyal, 6G Division of TEC)

9. Sh. Sudipta Bhaumik also emphasized that NWG-15 may submit a contribution on Indian experience on Telecom Infra Sharing for the next plenary session of Q7/SG15 in addition to reply to the questionnaire. He also shared the information that the SG15 has initiated to work

upon to revise the draft Recommendation ITU-T G.9964 regarding 'Unified high-speed wireline-based home networking transceivers – Power spectral density specification' and requested that the concerned experts from the NWG-15 may consider to prepare a contribution in this matter. He also highlighted that ITU has recently released the Technical report on "optical fibre, cable, and components for space-division multiplexing (SDM) transmission,". This technical report indicates a clear and agreed way forward for deploying and standardizing SDM optical fibre and cable technologies, with the goal of developing cost-effective SDM networks and the supporting business ecosystem. He also pointed out that India should work upon submitting contribution on FIFO and other emerging transport technologies.

10. Sh. Deepak Jain from IIT-Delhi informed that he is interested to work upon in the areas of Hollow-core fibre and SDM technologies and will submit a draft contribution in this regard at the earliest. Sh. Desai appreciated the initiative of Sh. Jain to come forward for the contributions and requested other members from Academia specifically to prepare contributions in their areas of interest and expertise. Sh. Desai also requested Sh. Jain to expedite the draft on the proposed contributions and share first draft for the consultation among the group by **24th June 2023**.

11. Sh. K. K. Minocha suggested to work upon the 'Life-cycle of OFC and its end disposal' keeping in view the circular economy initiatives at global level. For this, Sh. Desai and Sh. Bhaumik responded saying that environment and circular economy is within the scope of SG5/ NWG-5. Sh. Minocha agreed that he will pursue with NWG-5 in this regard.

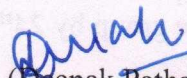
12. Ms. Sonali, HFCL has shown her interest in respect of contribution in FSOC technologies and related areas.

13. Sh. Desai requested to all the members to go through other possible areas of contributions and send their suggestions / draft contributions by **24th June 2023** so that the same can be discussed in next NWG meeting.

14. **The key action points emerged from the meeting are as under: -**

- i. The contributors of the work-item-FSOC (Q6/15) may prepare afresh contribution and also interact simultaneously with the Rapporteur of Q6/15 to explore the possibility of merging of Indian Contribution to the agreed work-item submitted by MIIT, China. Other members of the NWG-15 are also requested to give their inputs in the proposed work-item on FSOC. (Action By the contributors: Abdul Kayum/ Avadhesh Singh/ Rakesh Goyal / Rakesh Desai)
- ii. Indian Contribution on use cases of Li-fi in healthcare and education areas (Q3/15) may be updated further as and when required. (Action By the contributors: Rakesh Desai/ Mudita Chandra/ Hardik Soni, M/s NavTech / Deepak Solanki, Velminni)
- iii. Indian Contribution on "Topologies for optical access network" (Q7/15) may be updated further by the contributors at the time of finalization of the said new supplement L.Sup.fttx . Inputs are also invited from other members of the NWG-15 (Action By the contributors: Rakesh Desai/ Mudita Chandra/ BSNL).
- iv. Ms. Mudita, ADET, TEC under the guidance of Sh. Rakesh Desai is working on submitting Indian experience on outdoor optical infrastructures sharing. Members of the NWG-15 are requested to give their inputs by **15th July 2023** in respect of this work-item. (Action By: Rakesh Desai / Mudita)
- v. SG15 (Q5 to 8) has initiated discussions through correspondence activity to update of ITU-T Technical Paper LSTP-GLSR "Guide on the use of ITU-T L-series Recommendations related to optical technologies for outside plant". The NWG-15 members from Sterlite,

- HFCL, BSNL, DOT LSAs are requested to give their inputs by **13th June 2023**.
- vi. 6G Division of TEC may take a lead in preparing the contribution (Q3/15) regarding PQC/QKD implementation to make the system secure in the era of quantum computing. (Action By: Sh.Raju / Rakesh Goyal, 6G Division of TEC)
 - vii. Sh. Deepak Jain, IIT-Delhi may expedite the draft on the proposed contributions in the areas of Hollow-core fibre and SDM technologies and may share his first draft for the consultation among the group by **24th June 2023**.
 - viii. NWG-15 Members may please register on 'Standards Coordination Portal' of TEC (<https://tec.gov.in/scp/>) and start familiarizing with the features of the portal.
 - ix. Members may study all the questions of ITU-T SG15 and come forward to increase number of Indian contributions in ITU-T SG15.
15. The meeting ended with the vote of thanks to the chair.


(Deepak Pathak) 25/06/23
Convener, NWG-15 & Director (FA), TEC

To,
All the participants / Members of the NWG-15.

Copy to,
1. Sr.DDG & Head, TEC for kind information.

List of Participants

1. Sh. Rakesh Desai, DDG (FA) & Chairman/NWG-15, TEC
2. Sh. Abdul Kayum, DDG (J&K)
3. Sh. N Murali Krishna DDG SR, RTEC-SR, Bangalore
4. Sh. Deepak Jain, Professor, IIT, Delhi
5. Sh. Ravi Hegde, Professor, IIT Gandhinagar
6. Sh. Sudipta Bhuamik, Lead Technology, STL, Mumbai
7. Sh. K. K. Minocha,
8. Smt. Sonali, Sr. Manager (5G, BU, Standard & Research Group) HFCL
9. Sh. Rajeev Kumar Kaushik, BSNL
10. Sh. Arvind Kansal
11. Sh. Gyan Deep Verma, Post-Doctoral Fellow, IIT, Delhi
12. Sh. Rajeev Chaturvedi, Dura-Line India Ltd.
13. Sh. Deepak Pathak, Director (FA) and convener
14. Sh. Raj Mohan Meena, ADG (FN & Standard)
15. Sh. Naveen Khichar , ADET (TSOC), DoT HQ
16. Mrs. Mudita Chandra, ADET (Transmission), TEC
17. Sh. Vimal Kumar Singh, AD (FA), TEC
